TEST CODE **000592**

FORM TP 99111

MAY/JUNE 1999

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

MATHEMATICS

Paper 02 - General Proficiency

2 hours 40 minutes

03 JUNE 1999 (a.m.)

INSTRUCTIONS TO CANDIDATES

1. **Answer ALL questions in Section I, and any TWO in Section II.**
2. **Begin the answer for each question on a new page.**
3. **Full marks may not be awarded unless full working or explanation is shown with the answer.**
4. **Mathematical tables, formulae and graph paper are provided.**
5. **Silent electronic calculators may be used for this paper.**
6. **You are advised to use the first 10 minutes of the examination time to read through this paper. Writing may begin during this 10-minute period.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

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SECTION I

Answer ALL the questions in this section.

ALL working must be clearly shown.

1. (a) Calculate the exact value of

$$\frac{\frac{2}{3} + 1\frac{2}{7}}{3 - 2\frac{2}{7}}.$$

(3 marks)

(b) Evaluate $\frac{7.021}{6.751}$ and express the answer correct to

(i) 3 decimal places

(ii) 3 significant figures. **(3 marks)**

(c) Pamela borrows \$2 500 at 12% per annum simple interest for 2 years. In addition, a fee of \$200 is charged for processing the loan.

Calculate

(i) the sum of money, which she will have to repay the bank at the end of 2 years

(ii) the profit percent the bank makes on the loan

(iii) the monthly instalment, if she is required to repay the loan in 24 equal monthly instalments. **(6 marks)**

Total 12 marks

2. (a) Factorize completely:

(i) $y^2 - 3y$

(ii) $9x^2 - 1$

(iii) $8a^2 - 2a - 1$ **(4 marks)**

(b) Given that q varies directly as p, use the values of p and q in the table to calculate the values of a and of b.

p	2	8	a
q	6.1	b	1.2

(3 marks)

(c) 7 pencils and 5 erasers cost \$11.60, whereas 5 pencils and 3 erasers cost \$7.60. Calculate the cost of 8 erasers. **(4 marks)**

Total 11 marks

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3. (a) Given that $\varepsilon = \{a, b, c, d, e, f, g\}$ where ε is the universal set,

$L = \{a, b, c, d, e\}$

$M = \{a, c, e, g\}$

and $N = \{b, e, f, g\}$

(i) draw a Venn diagram showing the sets:
ε, L, M and N and their elements

(ii) list the members of the set represented by $(L \cap M) \cup N$

(iii) write down the value of $n((L \cap M) \cup N)'$. **(5 marks)**

(b) If $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} -3 \\ 1 \end{pmatrix}$, determine the values of a and of b. **(2 marks)**

(c) Given that vector $A = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$, calculate

(i) $|A|$, the length of vector A

(ii) the size of the angle made by vector A and the x-axis. **(3 marks)**

Total 10 marks

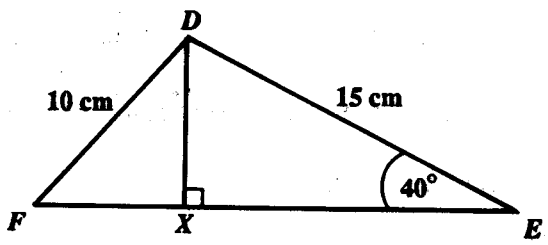
4. (a) (i) Using ruler and compasses only, construct ΔABC with $BC = 10$ cm, $AB = 4.5$ cm and $\angle ABC = 120°$. Bisect $\angle BAC$, such that the bisector of $\angle BAC$ meets BC at X.

(ii) Measure and state

– the length, in cm, of CX

– the size of the angle $A\hat{C}X$. **(5 marks)**

(b)



In the diagram above, **not drawn to scale**, DEF is a triangle with $DE = 15$ cm, $DF = 10$ cm and angle $DEF = 40°$. DX is perpendicular to EF.

Calculate

(i) the length, in cm, of DX

(ii) the size of the angle EDF. **(6 marks)**

Total 11 marks

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5. (a) If $h(x) = 1 + 3x$ and $k(x) = x + 2$,

Calculate

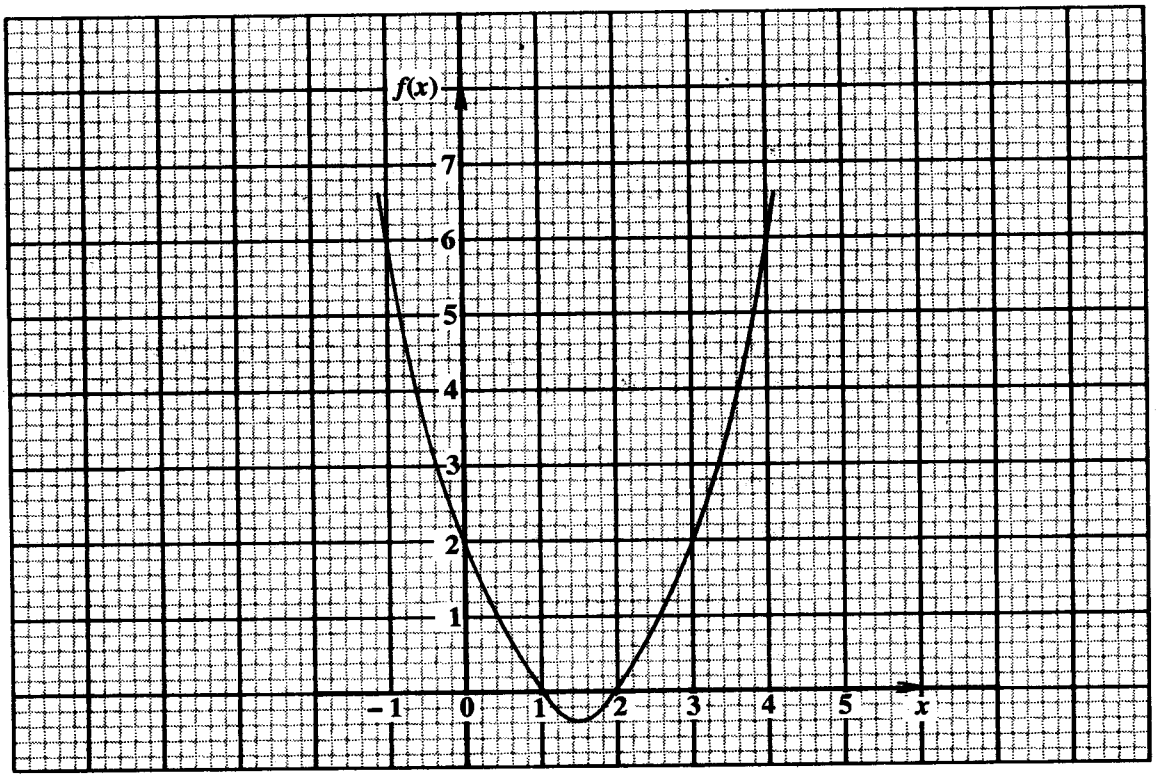
(i) $hk(x)$

(ii) $hk(4)$

(iii) $(hk)^{-1}(x)$

(iv) the value of x, when $hk(x) = 0$. **(5 marks)**

(b)



The graph shown above represents the function, $f(x) = x^2 - 3x + c$.

(i) Use the graph to determine

– the value of c

– the values of x for which $f(x) = 6$

(ii) Hence or otherwise determine the range of values of x for which

$x^2 - 3x + 2 < 6$. **(7 marks)**

Total 12 marks

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6. (a) The coordinates of the points L and N are (5, 6) and (8, – 2) respectively.

(i) State the coordinates of the midpoint, M, of the line LN.

(ii) Calculate the gradient of the line LN.

(iii) Determine the equation of the straight line which is perpendicular to LN and which passes through the point, M. **(6 marks)**

(b) (i) On graph paper, using a scale of 1 cm to represent 1 unit on each axis, draw triangle HIJ at $H(-2, 4)$, $I(-2, 1)$ and $J(-4, 1)$.

(ii) Given that T denotes a translation $\begin{pmatrix} -6 \\ -2 \end{pmatrix}$ and

R denotes a rotation of 180° with centre (1, 1),

draw the triangle $H''I''J''$, the image of triangle HIJ under the combined transformation TR, that is, rotation followed by translation.

(6 marks)

Total 12 marks

7. 38 pupils were asked to solve a problem, and the time taken, in minutes, by each of the pupils for the task was recorded.

The results are shown in the frequency table below.

Time (mins)	Frequency
6 – 10	2
11 – 15	7
16 – 20	12
21 – 25	15
26 – 30	2
31 – 35	0

(i) Calculate the mean number of minutes taken by pupils to solve the problem.

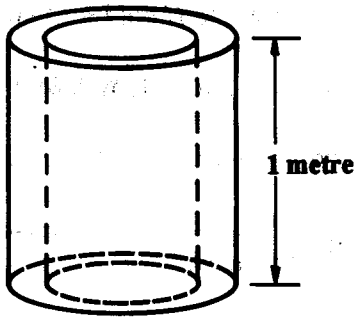
(ii) Draw a histogram to represent the data given in the table.

(iii) State a reason why the histogram, rather than the bar graph, is better suited to represent the data above.

(iv) Calculate the probability that a pupil, chosen at random, took more than 20 minutes to complete the problem.

Total 11 marks

8.



A circular drain pipe, shown in the diagram above, **not drawn to scale**, is 1 metre long, with outer and inner radii of 20 cm and 15 cm respectively.

(i) Draw a cross-sectional view of the drain pipe, showing the measurements of the outer and inner radii.

Calculate

(ii) the area, in cm^2, of the cross section of the drain pipe

(iii) the amount, in cm^3, of the material required to construct the drain pipe

(iv) the capacity, in litres, of the hollow space of the drain pipe

(v) the volume, in litres, of water passing through the pipe in 1 minute, if water flows through the pipe at a speed of 1 metre per second.

[Take π to be 3.14] **Total 11 marks**

SECTION II

Answer TWO questions in this section.

RELATIONS, FUNCTIONS AND GRAPHS

9. (a) (i) Express $2x^2 - 3x + 1$ in the form $2(x + p)^2 + q$ and state the values of p and q.

(ii) Hence, determine for $f(x) = 2x^2 - 3x + 1$

– the minimum value of $f(x)$

– the value of x for which $f(x)$ is a minimum. **(8 marks)**

(b) A minibus leaves town, X, at 09.00 hours to go to town, Y, at the constant rate of 60 km/h. At 09.30 hours, a car leaves Y to go to X, travelling at the constant rate of 90 km/hour. The distance between X and Y is 210 km.

(i) Using graph paper and taking 1 cm to represent 15 km on the distance axis and 1 cm to represent 30 minutes on the time axis, draw the distance-time graph showing the journeys of the minibus and the car.

(ii) From your graph, determine

– the distance from X when the vehicles meet

– the time when they meet. **(7 marks)**

Total 15 marks

10. (a) A firm uses a combination of large and small boxes to package the items it produces.

Large boxes can hold 8 items.
Small boxes can hold 3 items.

The firm wishes to package 84 or less items using no more than 18 boxes.

Let L represent the number of large boxes used and S, the number of small boxes used.

(i) Write down two inequalities, other than $L > O$ and $S > O$ to represent the information above.

(ii) Using a scale of 2 cm to represent 5 units on both axes, draw the graphs to represent the inequalities in (i) above and shade the UNWANTED region.

(iii) A profit of \$32 is made on a large box and \$8 on a small box.

Write down an expression to represent the total profit made on the boxes.

(iv) From your graph, determine the number of large boxes, L, and the number of small boxes, S, which will yield the maximum profit.

(v) Calculate the maximum profit.

Total 15 marks

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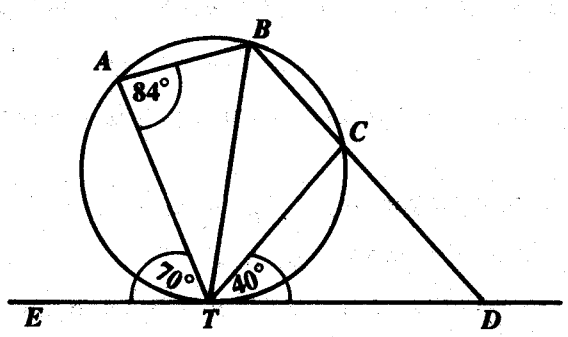
GEOMETRY AND TRIGONOMETRY

11. (a)



The diagram above, **not drawn to scale,** shows a quadrilateral, *ABCT*, inscribed in a circle. *DE* is a tangent to the circle at *T*.

Angle *ATE* = 70°, Angle *CTD* = 40° and Angle *TAB* = 84°.

Calculate, giving reasons for your answers, the magnitude of the angles:

(i) *ABT*

(ii) *BTC*

(iii) *CDT* **(6 marks)**

(b) Take the radius of the earth to be 6 400 km and π to be 3.14.

(i) Two towns, *Q* and *S*, are located at (40°N, 35°E) and (40°N, 145°W) respectively.

Calculate the distance, in km, between *Q* and *S* measured along their meridians and over the north pole.

(ii) *A* and *K* are two towns north of the equator, situated on the same circle of latitude. The distance between *A* and *K* is 3 612.8 km. *A* is on 20°E longitude and *K* is on 20°W longitude.

Calculate the latitude of *A* and *K*. **(9 marks)**

Total 15 marks

12. (a) Given $\tan\theta = \frac{12}{5}$,

calculate the exact value of

$\sin\theta + \cos\theta$. **(4 marks)**

(b) A ship leaves a point, C, and travels on a bearing of 037° to a point, D, 12 km from C. The ship changes its course and travels 25 km on a bearing of 140° to a point, E.

(i) Draw a sketch of the ship's journey, clearly marking in

- the north direction
- the bearing 037° and 140°
- the points C, D and E.

Calculate

(ii) the distance, in km, of CE to 2 decimal places

(iii) the bearing of E from C. **(11 marks)**

Total 15 marks

VECTORS AND MATRICES

13. (a) Two transformations are represented by

$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $M = \begin{pmatrix} -5 \\ 5 \end{pmatrix}$.

Write out a full geometrical description for P and for M. **(2 marks)**

(b) The transformation, Q, is represented by

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} -5 \\ 5 \end{pmatrix}$$

(i) Calculate the coordinates of

- the image S' of $S(-4, 3)$ under the transformation Q
- $L(a, b)$ which is mapped by Q onto $(-2, 1)$.

(ii) If $N(x, y)$ is mapped by Q onto (x, y), prove that ALL points on the line $y = x + 5$ are invariant points.

(iii) Describe geometrically, the transformation Q as a single transformation. **(13 marks)**

Total 15 marks

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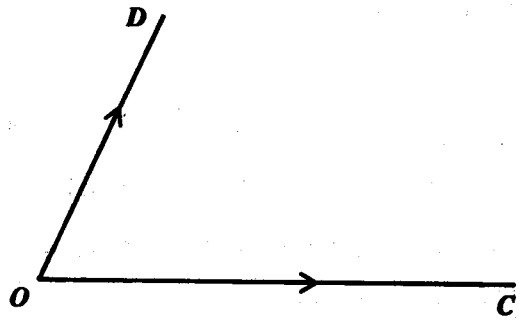
14.



(i) Copy and complete the diagram above to show:

$\overrightarrow{OD} = \underline{d}$

$\overrightarrow{OC} = \underline{c}$

X as the midpoint of OD

Y on $\overrightarrow{CD}$ such that $\overrightarrow{CY} = \frac{1}{4}\overrightarrow{CD}$

Z as the point of intersection of CX and OY

(ii) Express $\overrightarrow{CD}$, $\overrightarrow{CY}$ and $\overrightarrow{OY}$ in terms of $\underline{c}$ and $\underline{d}$.

(iii) Express $\overrightarrow{CX}$ in terms of $\underline{c}$ and $\underline{d}$.

(iv) Given that $\overrightarrow{CZ} = m\overrightarrow{CX}$, express $\overrightarrow{OZ}$ in terms of $\underline{c}$, $\underline{d}$ and m.

(v) If $\overrightarrow{OZ} = n\overrightarrow{OY}$, form an equation and determine the values of m and n.

Total 15 marks

END OF TEST